

Final M425 (I)

Gabriel Eze

May 2025

1 Introduction

Diffusion-weighted imaging (DWI) is a magnetic resonance imaging technique that is used to study the structure of white matter in the brain. However, methods such as diffusion tensor imaging (DTI) are limited—they only use a few gradient directions, which fails to capture complex fiber structures. To improve this, high-angular-resolution diffusion imaging (HARDI) methods have been developed, but they require long scan times.

This paper proposes a technique to enhance the angular resolution without additional scanning. By analyzing patterns in the raw HARDI signal, the method predicts new diffusion directions and synthesizes the corresponding signals using local signal similarity. These synthesized signals are then used to compute orientation distribution functions (ODFs) with improved accuracy.

2 Theory and Method

The core idea is to use known signals in measured diffusion gradient (DG) directions to estimate signals in unmeasured directions. These predictions rely on the idea that, within a local region of space, diffusion behaves similarly along neighboring directions.

For each unmeasured target direction, we then select a set of measured neighboring directions from the DG direction set $\mathcal{D}_N$. From this, triangles such as $\mathcal{N}_{\mathbf{v}_j}^{3,1} \subset \mathcal{D}_N$ are formed to define local geometric neighborhoods. The reliability of a triangle is assessed using correlation coefficients computed across a $3 \times 3 \times 3$ local window. These coefficients evaluate the similarity between the local diffusion patterns in the directions under consideration. If all pairwise correlations within a triangle are positive, it is deemed suitable for interpolation.

A new appropriate direction $\mathbf{u}_c$ is then determined as a weighted combination of the triangle's directions, lying within their convex hull:

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{jk}, \quad \text{where } \sum_{k=1}^K w_k = 1, \quad w_k \geq 0$$

Once the direction is predicted, the signal at $\mathbf{u}_c$ is synthesized by taking a corresponding weighted combination of the signals in the neighboring directions:

$$S(\mathbf{u}_c) = \sum_{k=1}^K w_k S(\mathbf{g}_{jk})$$

To extend the coverage of the signal over a finer angular grid, we then apply RBF interpolation. A Gaussian kernel is used to interpolate the signal at new directions using both measured and synthesized directions:

$$S(\mathbf{v}_j) = \sum_{g \in D_C} c_g \cdot \exp(-\alpha \|\mathbf{v}_j - g\|^2)$$

The interpolation weights $\mathbf{c}_g$ are solved from the linear system:

$$(A + \lambda I)\mathbf{c} = \mathbf{b}$$

3 Validation

To evaluate the effectiveness of the proposed interpolation method, the authors compare reconstructed spin distribution functions (SDFs) derived from three datasets:

1. f_{64} : Full measured signal (64 directions)
2. f_{32} : Partial measured signal (32 directions only)
3. f_{cm} : Partial signal + interpolated directions (32 + synthesized)

3.1 Relative L^2 Error

This is the primary quantitative metric:

$$RE_{32} = \frac{\|\tilde{f}_{64} - \tilde{f}_{32}\|}{\|\tilde{f}_{64}\|}, \quad RE_{cm} = \frac{\|\tilde{f}_{64} - \tilde{f}_{cm}\|}{\|\tilde{f}_{64}\|}$$

This measures how close the estimated SDFs are to the true distribution.

Crossing Fiber Voxels

The authors also calculate the percentage of crossing fibers that were detected in the brain from each datasets. Interpolated signals (f_{cm}) yield a higher percentage of accurately identified crossings than partial signals alone.

Angular Deviation and Fiber Ratio

The angular deviation between the estimated and true secondary fiber directions is calculated to assess directional fidelity. In Table 3 of the paper, the ratio of second fiber directions F_{d2} is reported for both f_{cm} and f_{32} relative to f_{64} , showing improved structural alignment using interpolated data.

Qualitative Comparison

Figure 4 of the paper displays voxel-wise visualizations of SDFs and their relative errors. These figures support the quantitative findings by highlighting spatial consistency improvements across the various data.

4 Results: Estimating Signals at Unmeasured Directions Using RBFs

The main goal of the method is to estimate diffusion signals at unmeasured gradient directions represented as **blue points** to increase angular resolution without resampling. These directions do not fall within the originally measured set or the directly predicted local directions, and require further interpolation based on surrounding information.

How RBFs Fill in the Gaps

The method estimates signal values in these blue directions by **blending nearby signals** using Radial Basis Functions (RBFs). Instead of a fixed neighborhood, RBFs consider all known directions which includes the measured and synthesized and compute contributions based on their angular proximity to the target.

- Each known signal contributes to the estimation based on how *close* its direction is to the desired target direction.
- The closer the direction, the higher the contribution since this is controlled by the Gaussian function shape.
- The result is a smooth, continuous approximation that respects the spatial variation of the diffusion signal across the sphere.

Why This Matters

- It allows interpolation over the entire spherical surface rather than just within local triangles.
- It handles non-linear signal variation, especially useful in areas with complex fiber crossings.
- It minimizes noise through regularization and smooth Gaussian decay.

Performance in Practice

- Lower reconstruction errors (RE_{cm}) compared to using partial signals alone.

- Better restoration of missing fiber orientations in both qualitative with the visual SDF shapes and quantitative metrics of the angular deviation, and fiber ratios.
- Improved consistency in tractography results, closely matching the output from fully sampled data.

5 Discussion

The use of RBFs in this setting is particularly effective because diffusion signals are known to vary smoothly with direction. The spherical Gaussian kernel exploits this fact, weighting contributions from nearby directions without relying on abrupt thresholds.

However, there are limitations:

- The interpolated signal at an unmeasured point cannot exceed the variation captured in the known data; peak values may be underestimated.
- The method performs best when the known directions are sufficiently dense and uniformly distributed.

6 Conclusion

This work demonstrates that radial basis function interpolation can be a powerful tool for estimating HARDI signals at unmeasured directions. By combining predicted intermediate directions and smooth interpolation across the sphere, the method provides:

- Enhanced angular resolution without additional scan time,
- Improved orientation distribution function (ODF) estimation,
- More accurate fiber reconstruction and tractography.

This framework can be applied to existing diffusion datasets and extended to multi-shell acquisitions or models like NODDI in future work.